Sample Paper 02

Class - 12th Exam - 2025 - 26

Mathematics (Code-041)

Time : 3 Hours **Max. Marks : 80**

General Instructions :

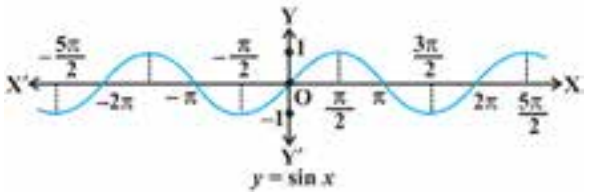
Read the following instructions very carefully and strictly follow them :

1. This Question paper contains 38 questions. All questions are compulsory.
2. This Question paper is divided into five Sections - A, B, C, D and E.
3. In Section A, Questions no. 1 to 18 are multiple choice questions (MCQs) and Questions no. 19 and 20 are Assertion-Reason based questions of 1 mark each.
4. In Section B, Questions no. 21 to 25 are Very Short Answer (VSA)-type questions, carrying 2 marks each.
5. In Section C, Questions no. 26 to 31 are Short Answer (SA)-type questions, carrying 3 marks each.
6. In Section D, Questions no. 32 to 35 are Long Answer (LA)-type questions, carrying 5 marks each.
7. In Section E, Questions no. 36 to 38 are Case study-based questions, carrying 4 marks each.
8. There is no overall choice. However, an internal choice has been provided some questions.
9. Use of calculators is not allowed.

SECTION - A

Section A consists of 20 questions of 1 mark each.

1. What is the maximum value of $y = \sin x$ observed in the graph?



(a) 0 (b) 1

(c) –1 (d) 2

2. The value of $\int_0^1 \frac{dx}{e^x + e}$ is

(a) $\frac{1}{e}\log\left(\frac{1+e}{2}\right)$ (b) $\log\left(\frac{1+e}{2}\right)$

(c) $\frac{1}{e}\log(1+e)$ (d) $\log\left(\frac{2}{1+e}\right)$

3. If $A = \{1, 2, 3, 4\}$ and $B = \{1, 2, 3, 4, 5, 6\}$ are two sets and function $f : A \to B$ is defined by $f(x) = x + 2$, $\forall\, x \in A$, then the function f is

(a) bijective

(b) onto

(c) one-one

(d) many-one

4. If $A = \begin{bmatrix} 3 & 4 \\ 5 & 7 \end{bmatrix}$, then $A \cdot (adj\ A)$ is equal to

(a) A (b) $|A|$

(c) $|A| \cdot I$ (d) None of these

5. The area of enclosed by $y = 3x - 5$, $y = 0$, $x = 3$ and $x = 5$ is

(a) 12 sq units (b) 13 sq units

(c) $13\frac{1}{2}$ sq units (d) 14 sq units

6. Solution of the equation $x\frac{dy}{dx} = y + x\tan\frac{y}{x}$ is

(a) $\sin\frac{x}{y} = cx$ (b) $\sin\frac{y}{x} = cx$

(c) $\sin\frac{x}{y} = cy$ (d) $\sin\frac{y}{x} = cy$

7. If $x = e^{y + e^{y + e^{y + \cdots}}}$, then $\frac{dy}{dx}$ is equal to

(a) $\frac{1}{x}$ (b) $\frac{1-x}{x}$

(c) $\frac{x}{1+x}$ (d) None of these

8. If A and B are two independent events such that $P(A) = \frac{1}{2}$ and $P(B) = \frac{1}{3}$, then P (neither A nor B) is equal to

(a) $\frac{2}{3}$ (b) $\frac{1}{6}$

(c) $\frac{5}{6}$ (d) $\frac{1}{3}$

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9. Objective function of a linear programming problem is

(a) a constraint

(b) a function to be optimized

(c) a relation between the variables

(d) none of the above

10. The point of intersection of the lines $\vec{r} \times \vec{a} = \vec{b} \times \vec{a}$ and $\vec{r} \times \vec{b} = \vec{a} \times \vec{b}$ is

(a) $\vec{a}$ (b) $\vec{b}$

(c) $\vec{a} + \vec{b}$ (d) $\vec{a} - \vec{b}$

11. The foot of the perpendicular from (0, 2, 3) to the line $\frac{x+3}{5} = \frac{y=1}{2} = \frac{z+4}{3}$ is

(a) $(-2, 3, 4)$ (b) $(2, -1, 3)$

(c) $(2, 3, -1)$ (d) $(3, 2, -1)$

12. Which of the following function is decreasing on $(0, \pi/2)$?

(a) $\sin 2x$ (b) $\cos 3x$

(c) $\tan x$ (d) $\cos 2x$

13. $2\sin^{-1}x = \sin^{-1}(2x\sqrt{1-x^2})$ holds good for all

(a) $|x| \leq 1$ (b) $1 \geq x \geq 0$

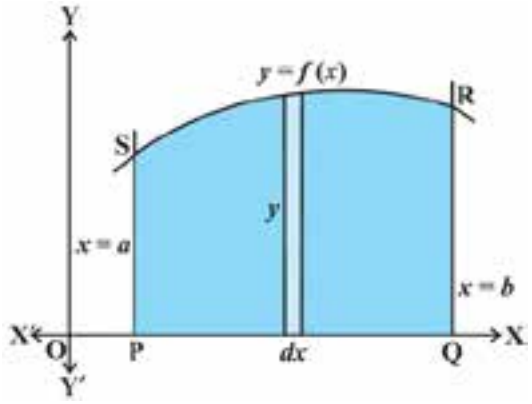
(c) $|x| \leq \frac{1}{\sqrt{2}}$ (d) none of these

14. If $f\left(x+\frac{1}{x}\right) = x^2 + \frac{1}{x^2}, x \neq 0$, then $f(x)$ is equal to

(a) $x^2 - 2$ (b) $x + 2$

(c) $x^2 + 2$ (d) $2x^2 - 5$

15. In Fig, what does the element labeled "dx" represent in the shaded region?



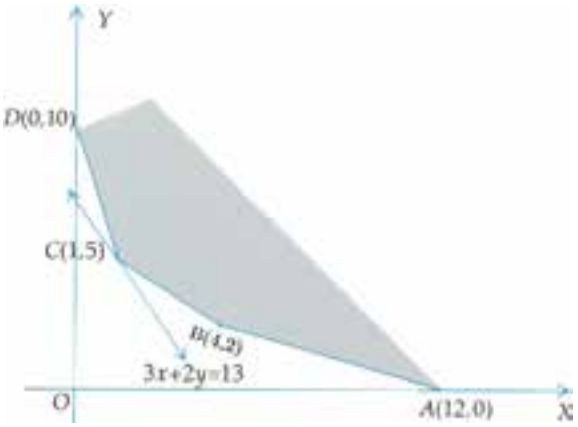
(a) The height of the rectangle under the curve

(b) The small change in the x-coordinate (width of the strip)

(c) The slope of the curve y=f(x) at a point

(d) The area under the curve between x=a and x=b

16. The minimum value of $z = 3x + 2y$, if the feasible region for an LPP is as shown in Fig, is to be determined.



(a) 36 (b) 16

(c) 20 (d) 13

17. The general solution of the differential equation $\frac{dy}{dx} + \sin\frac{x+y}{2} = \sin\frac{x-y}{2}$ is

(a) $\log\tan\left(\frac{y}{2}\right) = c - 2\sin x$

(b) $\log\tan\left(\frac{y}{4}\right) = c - 2\sin\frac{x}{2}$

(c) $\log\tan\left(\frac{y}{2} + \frac{\pi}{4}\right) = c - 2\sin x$

(d) $\log\tan\left(\frac{y}{4} + \frac{\pi}{4}\right) = c - 2\sin\left(\frac{x}{2}\right)$

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18. The degree of the differential equation satisfying $\sqrt{1-x^2}+\sqrt{1-y^2}=a(x-y)$

(a) 1 (b) 2

(c) 3 (d) 4

19. **Assertion:** $\int_0^{2\pi} \sin^3 x\, dx = 0$

Reason: $\sin^3 x$ is an odd function

(a) Assertion is true, Reason is true; Reason is a correct explanation for Assertion.

(b) Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

(c) Assertion is true; Reason is false.

(d) Assertion is false; Reason is true.

20. **Assertion:** The pair of lines given by $\vec{r}=\hat{i}-\hat{j}+\lambda(2\hat{i}+k)$ and $\vec{r}=2\hat{i}-k+\mu(\hat{i}+\hat{j}-\hat{k})$ intersect.

Reason: Two lines intersect each other, if they are not parallel and shortest distance $=0$.

(a) Assertion is true, Reason is true; Reason is a correct explanation for Assertion.

(b) Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

(c) Assertion is true; Reason is false.

(d) Assertion is false; Reason is true.

SECTION - B

This section comprises of very short answer-type questions (VSA) of 2 marks each.

21. Determine the value of k for which the following function is continuous at $x=3$:

$$f(x)=\begin{cases}\dfrac{(x+3)^2-36}{x-3}, & x\neq 3\\ k, & x=3\end{cases}$$

OR

Determine the value of constant k so that the function $f(x)=\begin{cases}\dfrac{kx}{|x|}, & \text{if } x<0\\ 3, & \text{if } x\geq 0\end{cases}$ is continuous at $x=0$.

22. If A and B are square matrices of the same order 3, such that $|A|=2$ and $AB=2I$. Write the values of $|B|$.

23. If $P(\text{not } A)=0.7$, $P(B)=0.7$ and $P(B/A)=0.5$, then find $P(A/B)$.

24. If $\vec{a}$ and $\vec{b}$ are perpendicular vectors, $|\vec{a}+\vec{b}|=13$ and $|\vec{a}|=5$, then find the value of $|\vec{b}|$.

25. Find the general solution of the following differential equation $(e^x+e^{-x})dy-(e^x-e^{-x})dx=0$

SECTION - C

This section comprises of short answer-type questions (SA) of 3 marks each.

26. Find the particular solution of the differential equation $(+e^{2x})\,dy+(1+y^2)\,e^x\,dx=0$, given that $y=1$, when $x=0$.

27. If $A=\begin{bmatrix}1&1&1\\1&0&2\\3&1&1\end{bmatrix}$, find A^{-1}.

Hence, solve the system of equations $x+y+z=6$, $x+2z=7$, $3x+y+z=12$.

OR

If $A=\begin{bmatrix}2&-3&5\\3&2&-4\\1&1&-2\end{bmatrix}$, find A^{-1}.

Use it to solve the system of equations

$2x-3y+5z=11.3x+2y-4z=-5, x+y-2z=-3.$

28. Find the coordinates of the foot of perpendicular drawn from the point $A(-1,8,4)$ to the line joining the points $B(0,-1,3)$ and $C(2,-3,-1)$. Hence, find the image of the point A in the line BC.

29. Evaluate $\int \frac{x^3+3x+4}{\sqrt{x}}$

30. Show that the relation S in the set R of real numbers defined as $S=\{(a,b): a,b\in R \text{ and } a\leq b^3\}$ is neither reflexive not symmetric not transitive.

31. If the function $f(x)$ given by $f(x)=\begin{cases}3ax+b, & \text{if } x>1\\ 11, & \text{if } x=1\\ 5ax-2b, & \text{if } x<1\end{cases}$ is continuous at $x=1$, then find the values of a and b.

OR

Find the value of k, so that the functions f defined by $f(x)=\begin{cases}kx+1, & \text{if } x\leq \pi\\ \cos x & \text{if } x>\pi\end{cases}$ is continuous at $x=\pi$.

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SECTION - D

This section comprises of long answer-type questions (LA) of 5 marks each.

32. Find the intervals in which the function given by $f(x) = \frac{3}{10}x^4 - \frac{4}{5}x^3 - 3x^2 + \frac{36x}{5} + 11$ is

(i) strictly increasing

(ii) strictly decreasing.

OR

A ladder 5 m long is leaning against a wall. Bottom of ladder is pulled along the ground away from wall at the rate of 2 m/s. How fast is the height on the wall decreasing, when the foot of ladder is 4 m away from the wall?

33. Maximise and minimise $Z = x + 2y$ subject to the constraints

$$x + 2y \geq 100$$
$$2x - y \leq 0$$
$$2x + y \leq 200$$
$$x,\ y \geq 0$$

Solve the above LPP graphically.

Corner points	$Z = x + 2y$
$A(0,50)$	$Z = 0 + 2 \times 50 = 100$
$B(20,40)$	$Z = 20 + 2 \times 40 = 100$
$C(50,100)$	$Z = 50 + 2 \times 100 = 250$
$D(0,200)$	$Z = 0 + 2 \times 200 = 400$

The maximum value of Z is 400 at $D(0,200)$ and the minimum value of Z is 100 at the points on the line segment joining $A(0,50)$ and $B(20,40)$.

OR

A manufacturer has employed 5 skilled men and 10 semi-skilled men and makes two models A and B of an article. The making of one item of model A requires 2 hours work by a skilled man and 2 hours by a semi-skilled man. One item of model B requires 1 hour by a skilled man and 3 hours by semi-skilled man. No man is expected to work more than 8 hours per day. The manufacturer profit on an item of model A is Rs. 15 and on an items of model B is Rs. 10. How many of items of models should be made per day in order to maximize daily profit? Formulate the above LPP and solve it graphically and find the maximum profit.

34. (i) Evaluate $\sin^{-1}\left(\sin\frac{3\pi}{5}\right)$.

(ii) Write the value of $\tan^{-1}\left(\tan\frac{3\pi}{4}\right)$.

(iii) Write the principal value of $\tan^{-1}\left[\sin\left(-\frac{\pi}{2}\right)\right]$.

OR

Using the principal value, evaluate the following:

(i) $\tan^{-1}1 + \sin^{-1}\left(-\frac{1}{2}\right)$

(ii) $\cos^{-1}\left(\cos\frac{2\pi}{3}\right) + \sin^{-1}\left(\sin\frac{2\pi}{3}\right)$

35. Find the shortest distance between the following lines. $\frac{x-3}{1} = \frac{y-5}{-2} = \frac{z-7}{1}$, $\frac{x+1}{7} = \frac{y+1}{-6} = \frac{z+1}{1}$.

OR

Solve graphically the following system of in equations.

$x + 2y \geq 20$, $3x + y \leq 15$

SECTION - E

Case study based questions are compulsory.

36. The Vande Bharat Express, also known as Train 18, is a semi-high-speed, intercity, electric multiple unit train operated by the Indian Railways on 4 routes as of October 2022. Routes include New Delhi to Shri Mata Vaishno

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In a survey at Vande Bharat Train, IRCTC asked the passenger to rate and review the food served in train. Suppose IRCTC asked 500 passenger selected at random to rate food according to price (low, medium, or high) and food (1, 2, 3, or 4 stars). The results of this survey are presented in the two-way, or contingency, table below. The numbers in this table represent frequencies. For example, in the third row and fourth column, 30 people rated the prices high and the food 4 stars. The last column contains the sum for each row, and similarly, the bottom row contains the sum for each column. These sums are often called marginal totals.

Price	**Food rating**			
Low	20	30	90	10
Medium	50	80	90	30
High	20	10	40	30

Assume that these results are representative of the entire passenger of train, so the relative frequency of occurrence is the true probability of the event. A passenger from train is randomly selected.

(i) Find the probability that the passenger rates the prices medium.

(ii) Find the probability that the passenger rates the food 2 stars.

(iii) Suppose the passenger selected rates the prices high. What is the probability that he rates the restaurants 1 star?

(iv) Suppose the passenger selected does not rate the food 4 stars. What is the probability that she rates the prices high?

37. Publishing is the activity of making information, literature, music, software and other content available to the public for sale or for free. Traditionally, the term refers to the creation and distribution of printed works, such as books, newspapers, and magazines.



NODIA Press is a such publishing house having two branch at Jaipur. In each branch there are three offices. In each office, there are 2 peons, 5 clerks and 3 typists. In one office of a branch, 5 salesmen are also working. In each office of other branch 2 head-clerks are also working. Using matrix notations find:

(i) the total number of posts of each kind in all the offices taken together in each branch.

(ii) the total number of posts of each kind in all the offices taken together from both branches.

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38. Different types of drugs affect your body in different ways, and the effects associated with drugs can vary from person to person. How a drug effects an individual is dependent on a variety of factors including body size, general health, the amount and strength of the drug, and whether any other drugs are in the system at the same time. It is important to remember that illegal drugs are not controlled substances, and therefore the quality and strength may differ from one batch to another.



The concentration $C(t)$ in milligrams per cubic centimeter (mg/cm^3) of a drug in a patient's bloodstream is $0.5\,\text{mg/cm}^3$ immediately after an injection and t minutes later is decreasing at the rate

$$C'(t) = \frac{-0.01e^{0.01t}}{(e^{0.01t}+1)^2}\,\text{mg/cm}^3 \text{ per minute}$$

A new injection is given when the concentration drops below $0.05\,\text{mg/cm}^3$.

(i) Find an expression for $C(t)$.

(ii) What is the concentration after 1 hour? After 3 hours?

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